CAN QUOTAS INCREASE THE SUPPLY OF CANDIDATES FOR HIGHER-LEVEL POSITIONS? EVIDENCE FROM LOCAL GOVERNMENT IN INDIA

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Abstract—A one-third quota rule for women in local political leadership seats in India increases the number of female candidates who later contest seats in state and national legislatures. This arises from the candidacy of beneficiaries who gained political experience due to the quotas and career politicians who continue contesting in longer-exposed areas. The policy accounts for a substantial portion of the increase in female candidates for high office since the mid-1990s. Women have a higher probability of a top finish when running on major party tickets or contesting in areas that overlap with their local constituency.

I. Introduction

QUOTAS are a common tool to increase the representation of historically underrepresented groups, and they are applied to varied sectors, including education, business, and politics. Often implicit in their advocacy is an argument that quotas can increase the participation of targeted groups in a way that can eventually render the policy obsolete through institutional change. This argument can be observed in effect when quotas are implemented in a specific segment of an organizational structure rather than uniformly, with the idea that the targeted segment will be sufficient to generate follow-on or spillover effects. Whether such institutional change is possible has long been debated in the many contexts in which quotas or affirmative action policies have been proposed and advocated (Coate & Loury, 1993).

In this paper, I ask, How do quotas for women in local elected bodies in India affect candidacy for and representation in higher offices? Myerson (2011) points to the primacy of a strong supply of capable local leaders for effective state building in a tiered and decentralized governance structure, and Acemoglu and Robinson (2012) make the case for inclusive institutions as a foundational tenet of economic development. The investigation that follows ultimately asks whether quotas intended to increase the inclusive nature of local political institutions can change the prospects for candidacy and representation of beneficiary groups in higher offices. I empirically test whether such a quota system can increase participation and representation in higher levels of government, and if so, through which channels. Whether electoral quotas can have broader effects in areas of governance to which they are not directly applied is an open question (Bhavnani, 2009; Iyer et al., 2012), and a burgeoning literature on the "personnel economics of the state" highlights the necessity of understanding the process by which individuals become candidates for public office (Finan, Olken, & Pande, 2015).

The Indian experience with quotas for women in local government provides an ideal environment to overcome several common empirical challenges. The first is identification. In implementing this policy, seat reservations were assigned across constituencies and over time so as not to be related to factors that otherwise might confound the election of female candidates with unobserved preferences. Second, I am able to estimate effects on margins of both candidacy and representation. Observing the set of candidates in other contexts where quotas are often applied can be difficult, as typically only representation is observed—despite candidacy being the margin on which representation gaps often first appear. Third, there is a sufficiently long study period available, allowing me to estimate longer-term effects, which should be a primary focus given the nature of the potential mechanisms for quotas to affect higher-level candidacy. Finally, by linking the same candidate across different elections to observe partial career histories, I am able to trace the channels through which candidates for higher office respond to the policy.

I show that longer exposure to quota-induced female leadership at the local level increases the number of female candidates for higher office among both state and national legislatures. State legislature constituencies that experienced ten additional years of exposure to female leaders fielded .15 more female candidates than constituencies with zero exposure; parliamentary constituencies fielded .41 more female candidates. Since state assembly constituencies are approximately one-tenth the size of parliamentary constituencies, the candidacy effect in state legislatures is approximately four times the effect among parliamentary constituencies when areas of the same size are considered, evidencing a career-ladder structure for local politicians choosing to contest elections in higher levels of government. The overall magnitude of the candidacy effect is substantive, implying that the local quota policy was responsible for around 50% of the increase in the number of female candidates running for office in higher levels of government in recent elections studied. These findings are robust across various specifications and alternative estimators, are strong in areas where local government constituencies overlap closely with state legislature and parliamentary constituencies, and effects are

Received for publication October 3, 2016. Revision accepted for publication July 9, 2018. Editor: Brigitte C. Madrian.

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I appreciate input from Onur Altindag, Ryan Brown, Ritam Chaurey, Andrew Friedson, Camilo Garcia-Jimeno, Mike Gechter, Chris Hanes, Lakshmi Iyer, David Jaeger, Francesca Jensenius, Barry Jones, Ted Joyce, Ben Olken, Rohini Pande, Pablo Querubin, Martin Rotemberg, Alex Segura, Yogesh Uppal, two anonymous reviewers, and seminar participants at Banco de la Republica, Binghamton University, CUNY Graduate Center, IFPRI, Louisiana State University, The New School, Oklahoma State University, Trinity University, and the University of Colorado–Denver. Partha Sarathi Bhowmick provided excellent research assistance. The views and conclusions expressed in this paper are my own.

A supplemental appendix is available online at http://www.mitpressjournals.org/doi/suppl/10.1162/rest_a_00802.

The Review of Economics and Statistics, March 2020, 102(1): 65–78
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https://doi.org/10.1162/rest_a_00802

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homogeneous across areas with varying existing socioeconomic and demographic characteristics.

By observing political career histories, I find the policy effect to be driven by two mutually exclusive groups of candidates. Approximately half of the increase in female parliamentary candidates is due to candidates who were likely beneficiaries of the quota system through previous service in local government. The remainder of the candidacy response is attributable to politicians who had previously contested, and lost, a state assembly election and continued contesting for higher office in areas that received longer quota exposure. Together, these findings suggest a combination of direct effects via cohorts of female politicians who contest elections at higher levels of government after gaining specific political experience provided by the quota system and indirect effects that make certain areas more conducive to continued female candidacy. Prior studies, notably Beaman et al. (2009), have established that voter attitudes change after exposure to female political leaders; this paper provides complementary evidence of a response among potential candidates to voters' exposure, as well as specific supply-side effects of quotas on later candidacy for higher offices.

I then investigate why female candidates do not have a higher probability of winning in areas with longer exposure. Overall, female candidates perform approximately as well as the median candidate in the elections they contest, but most of these new candidates run as independents rather than as part of major or minor parties. This suggests that new female candidates are not being incorporated into parties, resulting in (among other things) a lack of visibility and access to resources for campaigns at higher levels of government. There are substantive representation effects of quota exposure among elections in which female candidates run on major party tickets or contest in areas where the higher-office constituency closely overlaps with a single local government constituency. Effects among voter turnout are minimal, if they appear at all—disqualifying the hypothesis that quotas induce changes in the political system through broader enfranchisement and voter turnout.

The following sections review related literature and briefly discuss the history of seat quotas in local government in India. I then detail the data used and the empirical strategy, and present results for political participation and representation, along with a set of robustness analyses and investigations into mechanisms. The final section concludes.

II. Background and Literature

A. *Policy Background and Related Literature*

India has a federal system of government with direct elections for local, state, and national legislatures.[1] The country introduced nationwide seat quotas for women in government in 1993 with the 73rd and 74th Amendments to the Indian Constitution, which formalized a historical decentralized local governance structure known as the panchayat. The 73rd Constitutional Amendment Act instituted a three-tiered system of local government in rural areas consisting of, in increasing size, the village, subdistrict (block), and district levels. The amendments, which were intended to provide large-scale devolution and decentralization of powers to the local bodies, stipulated that members of the local governance bodies were to be elected at five-year intervals and provided for one-third of all seats at each governance level to be filled by women. These policies and their implementation have been the focus of a number of studies to date (further details can be found in Chattopadhyay & Duflo, 2004a, 2004b; Iyer et al., 2012; and Kalsi, 2017).

The 73rd Amendment stipulated that states had the responsibility to adjust or amend local elections to comply with its provisions. Most states eventually did so, often within a few years of the passing of the amendment (Iyer et al., 2012). The quota policy was applicable only to local governance, excluding state and national legislatures from having quotas for women. Once the provisions of the reform were implemented, one-third of seats were reserved for women at any level of the local governance hierarchy; for single-seat leadership positions, reservations were assigned randomly across areas in each election cycle such that in aggregate, the one-third quota would be met. This feature of rotating leadership assignment has been used to assess the effects of female leaders in previous studies, including Beaman et al. (2012) and Iyer et al. (2012). After several election cycles with the random assignment, there is considerable variation across areas in the cumulative number of years exposed to a woman in the leadership position; it is this variation in exposure to quotas applied to district leadership seats, at the highest level of this governance structure, that provides exogenous cross-sectional variation in exposure to female leaders used to identify dynamic cross-level effects in the political hierarchy.

This paper is most directly related to studies that investigate the effect of seat reservations in politics on the functioning of government and the welfare of constituents. One strand of this work assesses immediate effects on policy outcomes or changes outside the political system. In India, Chattopadhyay and Duflo (2004a) find differences in public good provision, while Beaman et al. (2012) find investments among young women attributable to changes in aspirations. Jensenius (2015), however, finds no effect of caste-based reservations on development outcomes in India over a period of the thirty years in which these quotas were active. In terms of institutional change, Iyer et al. (2012) find evidence that political empowerment resulted in greater reporting of crimes against women. Brollo and Troiano (2016) find evidence of reduced political patronage by female mayors in Brazil, while

[1]These three types of bodies are known as the panchayat, the state legislative assembly, and the parliament. The parliament consists of two houses, one of which is directly elected (Lok Sabha) and one appointed (Rajya Sabha). At the parliamentary level, I investigate effects on elections for the Lok Sabha, and all references to the parliament refer to this house.

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Ferreira and Gyourko (2014) find no change in the policies under female mayors compared to male mayors in the United States. Bardhan, Mookherjee, and Torrado (2010) find contrary evidence suggesting that quota-mandated female representation in India may have worsened the provision of services for some groups. Beaman et al. (2009) show how perceptions of women improve once men are exposed to women in leadership roles, and Khanna (forthcoming) highlights the importance of understanding changed incentives for those who are not directly eligible for affirmative action policies. In this paper, I also look at effects of quotas outside the particular environment to which they are applied. Whereas earlier work focused on the effects of quotas outside politics, I investigate follow-on effects within governance. Finally, the paper contributes to a new and developing literature investigating questions of personnel economics in the public sector (Finan et al., 2015) by asking whether temporary shocks can reduce the gender gap in representation among public servants via individual careers or institutional change.

Another strand of literature evaluates whether exposure to women in politics affects future candidacy via legacy effects of female political participation on both voters and candidates. While legacy effects have been found in a number of contexts for local politics, evidence is generally mixed, and effects on higher levels of government have rarely been investigated.[2] Bhalotra et al. (2018) find positive effects of a woman winning an election on subsequent female candidacy in state legislatures in India, which is largely due to the same candidates contesting again. Ongoing work finds that competitively won state legislature elections induces future female candidacy in parliamentary elections after the term of the elected legislator is completed (Brown, Mansour, & O'Connell, 2018). Broockman (2014) finds no effect among U.S. state legislature elections, and Uppal (2009) finds negative incumbency effects for men and women. Most closely related is Bhavnani (2009), who uses randomly assigned municipal council elections to determine the effects of seat reservation in municipal government in India on later candidacy in the same municipal councils, finding that quotas introduced women who continue running for office and demonstrated to the electorate that women could be viable candidates. Banerjee et al. (2017) use a similar approach to show that seat reservation had effects on incumbency and challenger entry. In this paper, I complement this literature by investigating the effects of female politicians across levels of government rather than within the same body.

This paper is also related to investigations of the effect of affirmative action and quotas in other sectors. Bertrand et al. (2019) study gender quotas for corporate boards in Norway and find that a mandated change in the gender composition of the upper echelons of firm governance yielded no effects on women outside these few positions in the same firm. In contrast, the Indian experience allows for an evaluation of the effect of quotas that would necessarily operate in the opposite direction to that studied by Bertrand et al. (2019) via the creation of a supply of experienced female candidates at a lower level of the organizational hierarchy. I next discuss the potential channels through which this relationship may be established in the context of electoral politics.

B. *Channels for Quotas to Affect Candidacy*

There are two channels by which exposure to quotas might increase the candidacy of women. A direct supply-side channel provides a cohort of particular women with experience in the political arena in local government, and some portion of these same women continue in a political career by contesting for higher office. The single district-level leadership seats are particularly germane to focus on as potential conduits for individual politicians to translate experience into candidacy for higher office: the district constituencies served by the chairperson consisted of areas (populations) that are nearly an order of magnitude larger than state assembly constituencies. The power accorded these leadership positions is thus substantial relative to those at lower levels of the local government and has the potential to accord both experience and exposure to the public at a level necessary for politicians to consider candidacy at higher levels of government.

The other channel is a demand-side mechanism in which exposure to female politicians changes voter attitudes and thus the viability of female candidates contesting elections in constituencies with greater exposure. (Empirical evidence for this channel has been established by Beaman et al., 2009, and Bhavnani, 2009.) In order to disentangle these channels, I augment typical election records by matching candidates over time and across levels of government to observe individuals' past candidacy and political career histories. Support for the candidate supply channel would be evidenced by the net policy effect being traced to individuals who served in local government due to quotas. While direct evidence for the demand-side effect would require voter-level information, response to changes in voter attitudes may be evidenced by candidacy response among women who were not previously involved in local government contesting in areas with greater exposure. While these two channels are not necessarily mutually exclusive, as politicians created by the quotas may have candidacy responses due to a combination of these factors, I show that both are present and contribute approximately equally to the net candidacy response to quota exposure.

III. Data

A. *Exposure to Female Leaders*

Data on the timing and location of reserved chairperson seats are available from Iyer et al. (2012), who digitized archival data from various sources for ten states' history of

[2]Studies of the legacy effects of female politicians on future candidacies in local politics include Nagarajan, Deininger, and Jin (2011) for India and Gilardi (2015) for Switzerland.

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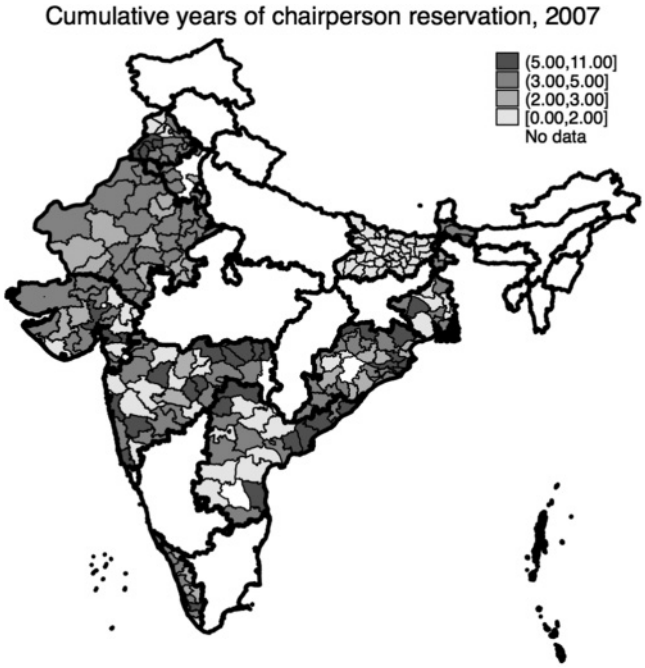
FIGURE 1.—POLICY VARIATION IN DISTRICT CHAIRPERSON EXPOSURE



Source: Author's calculations using data from Iyer et al. (2012).

TABLE 1.—AVERAGE YEARS OF EXPOSURE DISTRICT CHAIRPERSON QUOTAS

State	Mean	SD	Minimum	Maximum	*N*
Andhra Pradesh	4.1	2.6	0	11	22
Bihar	1.0	0.9	0	2	27
Gujarat	3.7	2.6	0	10	16
Haryana	4.3	0.9	3	5	9
Kerala	4.3	1.8	0	8	14
Maharashtra	4.1	2.2	0	8	28
Orissa	4.1	1.7	1	7	13
Punjab	3.1	2.9	0	8	10
Rajasthan	4.3	1.0	3	5	26
West Bengal	3.6	2.9	0	10	16
Overall sample	3.6	2.3	0	11	181

Source: Author's calculations using data from Iyer et al. (2012).

district-level seat reservations since the implementation of the 73rd Amendment.[3] Figure 1 shows the variation in exposure to district chairperson reservations as of 2007 for this sample, where districts that received more (less) exposure to the policy are more (less) heavily shaded.

Table 1 contains summary statistics of the cross-district variation in exposure to district chairperson reservations as of 2007 by state (as shown in figure 1). The mean level of exposure across states is relatively similar, but there is substantial within-state variation in years of exposure, reflecting the nature of the chairperson reservation assignments. Note, however, that some states (e.g., Haryana and Rajasthan) have little variation in exposure across districts while other states (including Gujarat, Andhra Pradesh, and West Bengal) have exposure ranging from 0 to 10 or 11 years.

Figure 2 depicts this same variation across states and districts. In this figure, each position on the vertical axis represents a separate district, with districts in the same states grouped together. For each district, the appearance of a filled line indicates that the district chairperson seat was reserved for a woman in that year (indicated on the horizontal axis). Districts that did not receive any period of reservation are indicated in gray within each state series. A few important patterns are worth highlighting. First, the assignment of chairperson seat reservation's are not perfectly rotating: at the end of each election cycle, some districts continue being reserved, while others switch status. Second, a number of districts received no reservation at all; these are present in seven of ten states. Third, Maharashtra followed a three-year election/reservation cycle, while all other states followed a five-year cycle. Additionally, there are two dimensions of exogenous variation arising after several election cycles: in accumulated exposure and in relative recency of exposure. In the analysis that follows, the variation in cumulative exposure will be used to identify net policy effects, while variation in recency will be used to explore whether the the effects of quota exposure increase or diminish over time.[4]

B. *Elections Data*

Data from parliamentary elections come from the Election Commission of India and contain the details of all candidates contesting in all constituencies in the 2009 election to the directly elected lower house of parliament (the Lok Sabha). Digitized historical state assembly election data, from Jensenius (2013), include vote counts and demographic information for candidates. Due to asynchronous elections across states, the cross-sectional sample is constructed from the set of elections held over the past four years for which the chairperson reservation data are available. These records are then associated with cumulative policy exposure as of the election year. The candidate data for both state assemblies and parliament are reported with a number of fields, most importantly candidate names, gender, vote shares, and constituency voter turnouts.

The candidate lists provide the potential to observe candidacy by the same individual across elections, which is used later to distinguish candidacy effects by candidates' career histories. Creating a viable candidate panel is not

[3]Comprehensive information on seat reservations is not readily accessible from any public sources or records. For their study, the authors of Iyer et al. (2012) collected data from diverse sources, including filing Right to Information Act requests for the history of district leader reservation assignments since the implementation of the 73rd Amendment. Some states did not respond to these requests, resulting in the ten-state sample shown in figure 1.

[4]Appendix table A1 tests means of demographic measures from the 1991 Population Census for districts in the ten-state sample versus all other states. Districts in states in the sample are larger in terms of population, have a higher rural female literacy rate and female educational attainment, a lower Scheduled Caste/Scheduled Tribe population share, and a higher female-to-male sex ratio. The ten-state sample consists of larger and more developed states (likely due to the nature of the data collection requiring historical records to be well kept and available). Heterogeneity analyses performed below find minimal evidence of effect heterogeneity across a wide set of area and population characteristics.

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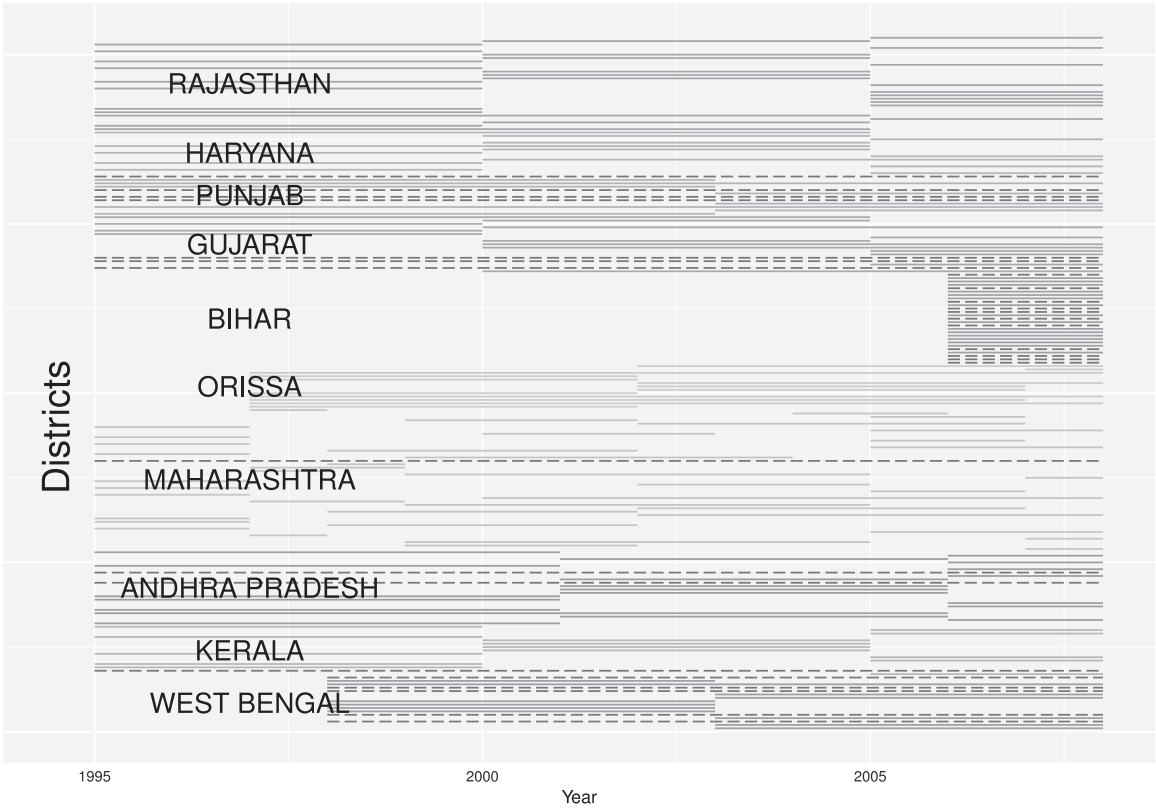
FIGURE 2.—VARIATION IN DISTRICT EXPOSURE TO SEAT RESERVATIONS: TIME LINES



Author's calculations using data from Iyer et al. (2012). In this figure, each position on the vertical axis represents a separate district, with districts in the same states grouped together. For each district, the appearance of a filled line indicates that the district chairperson seat was reserved for a woman in that year (indicated on the horizontal axis). Districts that did not receive any period of reservation are indicated by a dashed line within each state series.

straightforward, however, because there is no unique identifier field associated with candidates and a given individual's name may not be spelled or registered the same way across elections (due to differences in transliteration, honorifics, or name abbreviations, among others). In order to match candidates across elections, I apply an approximate string matching algorithm that searches for each candidate name in a given state and election with potential name matches in the following election cycle in the same state and with the same gender. Potential matches are then selected among names that are within a given string distance; next, they are reviewed manually, and a final match decision is made. Summary statistics can be found in appendix table A2, and the appendix contains a full description of this process.

C. *Matching Districts and Political Constituencies*

Parliamentary and assembly constituencies comprise the focal units of analysis given that elections are conducted according to political boundaries; district boundaries, which determine exposure to chairperson seat reservations, do not overlap with parliamentary or assembly constituencies (see appendix figure A1). To calculate an accurate measure of a constituency's exposure to female district leaders, I use GIS software to identify the intersection of the two sets of geographic areas in order to assign the respective portions of each district to its parliamentary constituency. This process then creates a set of unique geographic areas defined by the area pertaining to a unique combination of a district and constituency.[5] Table 2 provides summary statistics for these parliamentary and state assembly geographies.

IV. Analysis

A. *Identification*

Causal identification hinges on the exogeneity of the reservations assignment such that for the estimating equation,

$$Y_{s,d,c} = \delta_0 + \delta_1[Exposure_d] + \gamma_s + \mu_{s,d,c}, \quad (1)$$

where $Y_{s,d,c}$ is the outcome for state s in the unique area comprising the overlapping area of district d and constituency c, $Exposure_d$ measures the number of years that the district leadership seat had been reserved for a woman up to 2007, and γ represents a vector of state fixed effects. The

[5]This intersection process results in the geographic units used in the analysis. If districts are indexed by d and constituencies (either parliamentary or assembly) by c, a unit thus has a particular definition as the area that comprises the unique intersection of district d and constituency c, the area defined by d, c.

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TABLE 2.—SUMMARY STATISTICS FOR INTERSECTED CONSTITUENCY RETURNS DATA SETS

Variable	Mean	SD	Minimum	Maximum
A. Assembly constituencies				
Candidates	8.237	4.073	2	33
Number of female candidates	0.414	0.666	0	4
Whether area had any female candidate	0.330	0.470	0	1
Whether female candidate won election	0.066	0.248	0	1
Vote share for female candidates	0.069	0.163	0	0.973
Years exposure to chairperson reservation	2.715	2.504	0	10
B. Parliamentary constituencies				
Candidates	13.495	5.911	4	38
Number of female candidates	0.857	1.087	0	5
Whether area had any female candidate	0.520	0.500	0	1
Whether female candidate won election	0.109	0.312	0	1
Vote share for female candidates	0.085	0.175	0	.904
Years exposure to chairperson reservation	3.396	2.315	0	11

Summary statistics reflect unconditional means across parliamentary and state assembly constituencies intersected with district boundaries in the ten-state sample shown in table 1. Panel A is based on 1,615 constituencies intersected with 181 districts, resulting in 2,995 observations. Panel B is based on 278 constituencies across 181 districts, resulting in 1,375 intersecting geographical units.

residual is orthogonal to exposure, conditional on γ, that is, $\text{cov}(\mu_{s,d,c}, Exposure_d|\gamma_s) = 0$.

Identifying variation arises from two aspects of the policy. First, some areas were reserved for multiple election cycles within the first fifteen years of implementation, while others were reserved only once or not at all. Second, some areas had not yet completed their reserved term of office as of the focal assembly or parliamentary election simply because they were assigned reservation in a later election cycle than other areas. The following section provides evidence that the variation in exposure is not correlated with observable characteristics of constituencies. That is, substantial evidence supports the proper implementation of the random assignment of the leadership seat reservations across constituencies and election cycles.

B. *Falsification Tests for Selection on Observables*

Figure 3 provides visual evidence of exogeneity in exposure to female leaders in district chairperson seats by plotting an area's eventual exposure level against a prepolicy measure of female political participation (the share of female candidates in the election prior to 1993). In the top panel, I show this relationship for state assembly constituencies: the left-hand graph shows the unconditional relationship, and the right-hand graph shows the relationship after removing state fixed effects (i.e., the within-state variation shown in figure 1 that will be used to identify effects as described in the empirical strategy). The bottom panel follows a similar pattern for parliamentary elections. In both cases, a flat line is evidence against a systematic relationship between cumulative exposure and preexisting differences in the propensity for women to be involved in elections for state and national legislatures.

For the analysis that follows, I focus on the relationship between policy exposure and constituency-level measures of female and male candidacy. To formalize the above falsification test, I collect a set of prepolicy measures of district socioeconomic characteristics from various sources and use them to predict the election outcomes that are the focus of analysis. This predicted outcome is then regressed on the policy measure to test whether this aggregate of prepolicy local characteristics encapsulated in the predicted outcome can be explained by later policy variation. These characteristics are taken from the 1991 population census, India's household labor force survey carried out by the National Sample Survey Organization in 1987–1988, and election records for the 1991 general election. The indicators include demographic characteristics (average household size, sex ratios, ethnic mix), school enrollment rates, female literacy rates, household consumption per capita, and prepolicy measures of female political participation.

Table 3 presents these estimations. For both state and national legislatures, the policy measure has trivial explanatory power for the predicted outcomes. Another test lending support to exogeneity in the policy measure across area observables is a direct test of the degree to which prepolicy area characteristics can predict the policy measure. In appendix table A3, I show that an F-test of the vector of coefficients on the prepolicy area characteristics fails to reject the null hypothesis at any traditional level of significance. Taken together, these tests suggest strong within-state balance in prepolicy demographic, social, and political characteristics across districts receiving different eventual exposure intensities.

C. *Estimation*

The empirical strategy is straightforward. I regress a focal measure of political participation on the accumulated exposure to female district leaders conditional on a vector of state fixed effects. Regressions are weighted by the population share in the constituency based on data from the Gridded Population of the World (CIESIN, 2005) matched to the corresponding geographical area. By construction, the error terms are correlated by district (due to the policy variation) and constituency (the units for which elections outcomes are observable), so standard errors are two-way clustered by district and constituency.

Table 4 contains point estimates for the number of female and male candidates when separately regressed on total years of chairperson reservations for state assembly elections. The first column of table 4 shows that an additional year of seat reservation increases the number of female candidates for the state legislature by a small but statistically distinguishable .015. That is, for a constituency that experienced

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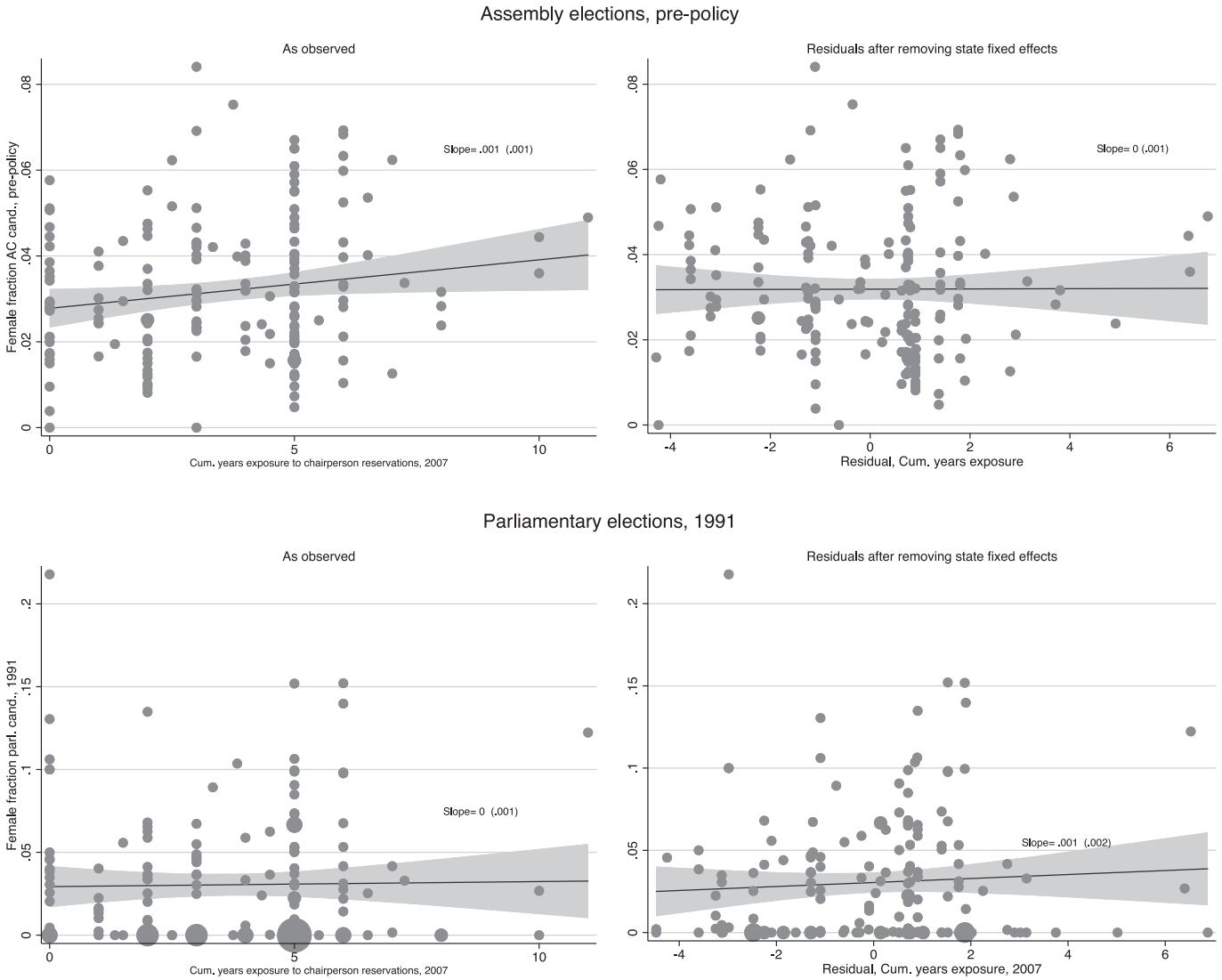
FIGURE 3.—FALSIFICATION: PREDICTING PREPOLICY FEMALE POLITICAL PARTICIPATION WITH CUMULATIVE EXPOSURE



These figures show the unconditional (left) and conditional (right) relationship between prepolicy female political participation as measured through state and national legislature candidacy rates and years of exposure to women leaders via seat quotas from 1995 to 2007. A flat line indicates no relationship between prepolicy measures and eventual exposure.

two election cycles (ten years) of reservations relative to a constituency receiving no exposure, the former would expect to see .15 (or approximately .22 standard deviations) more female candidates in this higher-level election. While there is a negative sign on the number of male candidates, this effect is not estimated precisely (column 2).

Because districts do not always overlap with political constituencies, we may be concerned about measurement error induced by the fact that candidates typically run in their home constituency. Say that district A is bifurcated evenly by a parliamentary constituency boundary creating subareas A1 and A2. If district A received sufficient policy exposure to generate an additional candidate, that candidate is likely to contest future elections only in her home constituency. Policy exposure in the other area may therefore be mismeasured, given that the exposure was not accruing to a potential candidate within its boundaries. Since this measurement error can only be positive, it is likely to lead to downward bias.

Although we cannot observe candidate residence, we can instead limit the sample to heavily overlapping areas in which this type of mismeasurement is less likely to be the case. In these cases, we are more sure that these areas will be treated in the sense that potential candidates are likely to live in and later contest from the constituencies they fall in. In columns 3 and 4, we limit the sample to constituency areas that overlap more than 80% of their population with the underlying district.[6] This sample restriction only slightly increases the effect magnitudes among state legislature elections, which is

[6]Note that because of the relative size of state assembly constituencies and parliamentary constituencies, we expect this mismeasurement to be an issue primarily among parliamentary elections.

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TABLE 3.—TESTING POLICY VARIATION IN EXPLAINING PREPOLICY PREDICTED OUTCOMES

	Outcome: Covariate-Predicted Measures of Candidacy			
	State Legislatures		Parliament	
	Female Candidates (1)	Male Candidates (2)	Female Candidates (3)	Male Candidates (4)
Years reserved	0.002	−0.056	0.007	0.007
	(0.003)	(0.037)	(0.005)	(0.032)
Mean of outcome	0.45	7.60	0.90	12.50
SD of outcome	0.125	1.66	0.28	3.52
N	2,995		1,375	

This table reports coefficient estimates of the effect of an additional year of exposure to female leaders on measures of candidacy as predicted by prepolicy observables. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

TABLE 4.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON STATE LEGISLATURE CANDIDACY

	Full Sample		Heavily Overlapping Sample	
	Female Candidates (1)	Male Candidates (2)	Female Candidates (3)	Male Candidates (4)
A. Unconditional				
Years reserved	0.015*	−0.054	0.016*	−0.057
	(0.009)	(0.058)	(0.009)	(0.060)
B. Including covariates				
Years reserved	0.014*	−0.051	0.016*	−0.054
	(0.008)	(0.057)	(0.008)	(0.059)
Mean of outcome	0.41	7.82	0.44	7.76
SD of outcome	0.66	4.13	0.68	4.23
N	2,995		1,409	

This table reports coefficient estimates of the effect of an additional year of reservation on constituency-level metrics of candidacy. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency are in parentheses. Significant at * <.1, ** <.05, and *** <.01.

expected due to the fact the most state legislature constituencies are uniquely contained within districts. There are, however, sizable differences when this is applied to parliamentary elections.

Table 5 contains estimates of candidacy measures in parliamentary elections. Coefficient patterns are similar to effects among assembly elections, although the candidacy effect is substantially larger: for an additional ten years of exposure to female district leaders, there are an additional .41 female candidates (or approximately .4 standard deviation). At the mean level of exposure, almost one in six districts fielded an additional female candidate ($0.041 \times 3.4 = .14$ additional female candidates per constituency). There is again no evidence of a response among male candidates in either direction. For both state legislatures and parliamentary elections, table 5B shows that results are similar in pattern and magnitude when controlling for prepolicy factors, including prepolicy female candidate shares and local demographic characteristics. When limiting to the sample of heavily overlapping area, the estimates increase in both magnitude and significance: unadjusted estimates in panel A suggest an additional year of exposure increases female candidacy by 0.065 (50% larger than in the attenuated estimates), while adjusted estimates in panel B are around 25% larger. Appendix figure A2 shows the evolution of this effect across the distribution of sample restrictions.[7]

TABLE 5.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON PARLIAMENTARY CANDIDACY

	Full Sample		Heavily Overlapping Sample	
	Female Candidates (1)	Male Candidates (2)	Female Candidates (3)	Male Candidates (4)
A. Unconditional				
Years reserved	0.041*	0.113	0.065**	0.106
	(0.021)	(0.109)	(0.027)	(0.148)
B. Including covariates				
Years reserved	0.035	0.109	0.053*	0.080
	(0.022)	(0.103)	(0.030)	(0.142)
Mean of outcome	0.86	12.64	0.89	12.48
SD of outcome	1.08	5.64	1.05	5.55
N	1,375		176	

This table reports coefficient estimates of the effect of an additional year of reservation on constituency-level metrics of candidacy. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency are in parentheses. Significant at * <.1, ** <.05, and *** <.01.

The directional correspondence in effects of exposure to female leaders across two levels of government confirms that district leadership seats have a potentially pivotal role in according experience that is sufficient to enhance the viability of a candidate in other elected bodies. It is also important to note an average of nine state legislature constituencies per district; scaling the coefficient then allows us to see that an additional two election cycles of exposure generate between one and two additional female candidates for the state legislature, compared to around 0.5 candidates for the parliament for a similar level of exposure and comparable area. That the effect is stronger at the state legislature level when comparing similar areas also suggests that the state legislature may be seen as a logical intermediate career step for politicians from previously reserved areas. The effect magnitude also very closely mirrors the number of district chairpersons who would have been available for higher office candidacy, with ten years, or two cycles, yielding approximately two new female politicians available to run for either office.

The estimated magnitudes of the policy effect are substantial in view of the total increase in the number of female candidates for office in these elections. There were 157 female candidates in these states in the 1991 parliamentary elections

[7]Appendix figure A2 first plots coefficient magnitudes and *t*-statistics for the whole area sample, corresponding to estimates in tables 4 and 5. I then remove units with small constituency area shares from the sample and reestimate the specification. The coefficient magnitudes and *t*-statistics are plotted, according the area share cutoff along the horizontal axis. The exercise begins with a cutoff of 0 and is then reestimated for samples that increase the threshold until only areas that comprise 80% or greater of the total constituency area are left. For reference, the unique constituencies comprising the restricted sample are plotted at intervals of ten at the top of the graph. This leaves a sample of constituencies that cleanly overlap with a single district comprising the majority of their area, with the corresponding estimates plotted to the right of the graph.

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TABLE 6.—NONLINEARITIES IN EFFECTS ON FEMALE CANDIDACY

Interaction:	(1)	Exposure (2)	Literacy (3)	Average Years of Education (4)	Sex Ratio (5)	Low-Caste Population Share (6)
A. State assemblies						
Years reserved	0.015*	−0.026	0.016*	0.015*	0.016*	0.014*
	(0.009)	(0.019)	(0.009)	(0.009)	(0.009)	(0.008)
Interaction		0.056**	0.053	0.026	0.051	0.117
		(0.026)	(0.066)	(0.060)	(0.055)	(0.082)
B. Parliament						
Years reserved	0.041*	0.049	0.043**	0.043**	0.041*	0.041*
	(0.021)	(0.056)	(0.020)	(0.020)	(0.021)	(0.021)
Interaction		−0.009	−0.090	−0.253	−0.045	0.035
		(0.055)	(0.206)	(0.243)	(0.183)	(0.245)

This table reports coefficient estimates of the effect of an additional year of reservation on the number of female candidates by constituency. Coefficients are from the estimation of equation (1), including an additional interaction term with the measure indicated in the column heads. Measures of local characteristics used in columns 3–6 are unit-standardized, and main effects of the interaction terms are unreported. Estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Panel A estimates are based on a sample of 2,995 observations; panel B estimates are based on a sample of 1,375 observations. Heteroskedasticity-consistent standard errors clustered by district and constituency are in parentheses. Significant at * <.1, ** <.05, and *** <.01.

compared to 265 in the 2009 elections. With an average of 3.4 years of exposure to the policy across 278 constituencies, the point estimate of 0.041 implies an additional 38.75 female candidates as a result of the policy, comprising approximately 35% of the increase in female candidacy seen in these states over this period. In state legislatures, the policy effect is smaller in magnitude, but it is similarly responsible for a large share of the increase in female candidates: with an average exposure of 2.8 years across 1,615 constituencies, the point estimate of 0.015 implies an additional 67.8 female candidates running for state legislatures—the majority of the increase from 705 to 802 female assembly candidates in these states from before the policy to the most recent election.

V. Robustness and Effect Heterogeneity

A. *Nonlinearities and Moderators*

Table 6 interacts the exposure measure with various local demographic and socioeconomic characteristics to explore whether the overall effects on candidacy are concentrated in areas with any particular characteristics.[8] Column 1 repeats the unconditional OLS specifications from tables 4 and 5 in panels A and B, respectively. Column 2 reports results from a specification including the square of the exposure measure, while columns 3 to 6 interact the exposure measure with rural female literacy rates and educational attainment, the population sex ratio, and the population share of lower-caste groups. Overall, the estimates of the main effect of exposure are remarkably robust across these specifications. One exception is the polynomial specification (column 2), in which effects are imprecisely estimated for parliamentary elections and the squared term becomes more prominent among the state assembly elections. Candidacy results are robust to the use of count data (Poisson) models in appendix table A4.

B. *Sample Construction and Composition*

An alternative to the current sample construction method would be to calculate the weighted average policy exposure for each constituency based on the various district components comprising the constituency. This also obviates the need for multiway clustering of standard errors, although the resulting tests would be less conservative in rejecting the null due to uncorrected partial error term correlations. Appendix table A5 shows these estimates for both the state assembly and parliamentary elections, with results mirroring (or stronger than) those in tables 4 and 5. Results are also robust to weighting observations by constituency area share instead of population, as seen in appendix table A6.

VI. Causal Channels for the Candidacy Effect

A. *Prior Exposure to Politics*

The results above suggest that exposure to female leaders via quotas at a lower level of the political hierarchy increases the political candidacy decisions of women running for higher office. To better understand the ways in which the policy affects the candidacy margin, I investigate characteristics of the candidates who are revealed to be moved by the policy. To do this, I first use the panel nature of the candidate data set to incorporate historical information on candidacy and public service to determine if the effects in table 5 are particularly concentrated among either repeat candidates or incumbents.

I segment candidates by various measures of prior political experience and exposure, with the focal outcomes now defined as the number of female candidates who are also (a) candidates with prior exposure in local government, (b) repeat (parliamentary) candidates, (c) previous candidates for lower office, and (d) candidates with known family involvement in politics (Labonne, Parsa, & Querubin, 2017). Table 7

[8]This investigation also allows the ability to extrapolate about the generalizability of effects in the whole of India beyond the ten-state sample available for the analysis. Appendix table A1 contains results from a test across samples in the same indicators used below, showing differences across in-sample and out-of-sample districts in their population size, female literacy rates and educational attainment, and female-to-male sex ratio.

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TABLE 7.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON PARLIAMENTARY CANDIDACY VIA CAREER POLITICIANS ($N = 1{,}375$)

Effect channel:	*Candidate Supply*	*Response to Voter Demand*		*Other*
Measure:	Prior Local Politician (1)	Female Previous Parliamentary Candidate (2)	Female Previous State Legislature Candidate (3)	Known Political Family (4)
Years reserved	0.015*	0.003	0.026***	0.016*
	(0.007)	(0.002)	(0.009)	(0.009)
Mean of outcome	0.10	0.02	0.19	0.14
SD of outcome	0.32	0.15	0.39	0.39

This table reports coefficient estimates of the effect of an additional year of reservation on constituency-level metrics of within-level and cross-level repeat candidacies. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

presents point estimates when estimating the same cross-sectional specification as above.

By observing partial career histories of individual candidates, we see that the policy effect among female candidates for parliament is due to two groups of women: those who had previously served in local government (column 1) and those who had previously contested (but not won) state assembly elections and continued in politics to contest parliamentary elections (column 3). These two groups are almost entirely distinct (shown in appendix table A7). The third category, comprising all other female candidates about whom no background information was found and in which new female candidates would be captured, does not exhibit any response to the policy exposure. The simplest explanation for this would be that policy exposure lowers the cost of candidacy only enough to affect those who have already incurred the fixed costs of establishing a political career—those having previously contested or served. These results are important for a number of reasons. First, the effects found at higher levels of the political organization come, at least partially, from individuals with repeat candidacies. Short-run effects arising through this channel are thus likely to be small, highlighting the need to examine these dynamics from a cumulative, longer-term perspective. Second, the bottom-up quota policy affects other levels of the organizational hierarchy at least in part thorough internal candidates, some of whom were originally introduced into politics through quotas. This stands in particular contrast to the lack of effects outside the upper echelons of corporate structures from the top-down quota policies for women in several European countries (Bertrand et al., 2019). Finally, this is evidence for both the candidate-supply mechanism (being the same individuals who gained experience in local government now contesting) and candidate response to changes in voter demand, where prior higher-office candidates continued contesting elections in areas that had historically been exposed to women politicians in positions of particular power. While changes in attitudes after exposure to female leaders have been found previously (Beaman et al., 2009), table 7 provides evidence suggesting a response among candidates who were not direct beneficiaries of the quota policy themselves—potentially due to broader institutional changes in attitudes toward women as effective leaders and viable candidates in constituencies with greater past exposure to female leaders.

B. *Recent versus Past Exposure and Exposure Intensity*

I next separate variation in the main exposure measure into periods of recency relative to the focal election. If the effect of female leaders on candidacy was through some measure of salience or inspiration (either directly or through expectations of immediately enhanced voter support or turnout), we might expect to see effect sizes increasing in recency; if the primary channel by which candidacy is affected is through providing opportunities for political experience, effect sizes may be less tied to particularly recent experience with quota exposure.

For this analysis, the policy measure is separated into three periods of exposure recency relative to the focal election: exposure in the preceding four years, exposure five to nine years prior, and early exposure in the first four years of the policy. These cutoffs are created in order to capture the fact that terms of office are for five years; this ensures, for example, that an area experiencing reservation five to nine years prior would necessarily have had a completed term of office for those in reserved seats at the time of the focal election. Estimates for state assembly elections are in columns 1 and 2 of table 8; effects in parliamentary elections are in columns 3 and 4. Across both levels of government, it is clear that the main results from tables 4 and 5 are largely driven by exposure in the moderate past, in which the particular leaders who would have been in the chairperson seat would have had time to complete their full term of office and used that experience to establish a foundation for or begin their next political campaign. For parliamentary elections, it is clear that short-term effects are small, and the most meaningful exposure comes from a moderately recent exposure period. This lack of effect among the most recent exposure category also refutes the hypothesis that female candidates run because there will be better cross-level support from those currently in office in local government via the quota system. In the parliament, it is clear also that effects of exposure decrease after moderate recency, suggesting an impermanence in the effects, further supporting the hypothesis that the effect of quotas on later candidacies is concentrated specifically among those who gain political experience as a result of the quotas that can be drawn on in soon-upcoming political campaigns. One can also segment the total amount of exposure to the policy into year ranges to gauge nonlinearity in effects. In table 9, I split the policy exposure measure into four ranges: one to two

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TABLE 8.—SEPARATING EFFECTS OF RECENT AND PAST EXPOSURE

	State Legislatures		Parliament	
	Female Candidate (1)	Male Candidate (2)	Female Candidate (3)	Male Candidate (4)
Years reserved, 10–15 years prior	0.022	0.070	−0.031	−0.003
	(0.027)	(0.139)	(0.038)	(0.198)
Years reserved, 5–9 years prior	0.021	−0.125	0.091**	0.168
	(0.017)	(0.105)	(0.036)	(0.172)
Years reserved, 0–4 years prior	0.003	−0.028	0.012	0.150
	(0.016)	(0.104)	(0.045)	(0.234)
Mean of outcome	0.41	7.82	0.86	12.64
SD of outcome	0.66	4.13	1.08	5.64
N	2,995		1,375	

This table reports coefficient estimates of the effect of an additional year of reservation in different periods of recency on constituency-level metrics of candidacy and election outcomes. Coefficients are from the estimation of this modified form of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

TABLE 9.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON STATE AND PARLIAMENTARY ELECTION OUTCOMES: HETEROGENEITY BY EXPOSURE INTENSITY.

	(1) State (Full Sample)	(2) State (Overlapping Sample)	(3) Parliament (Full Sample)	(4) Parliament (Overlapping Sample)
1–2 years reserved [0/1]	−0.167**	−0.165**	0.256	0.235
	(0.069)	(0.075)	(0.201)	(0.275)
3–5 years reserved [0/1]	−0.009	−0.017	0.222	0.220
	(0.046)	(0.047)	(0.159)	(0.196)
6–7 years reserved [0/1]	−0.002	0.010	0.626***	0.806**
	(0.075)	(0.081)	(0.230)	(0.335)
8 or more years reserved [0/1]	0.224**	0.245**	0.286	0.670**
	(0.110)	(0.119)	(0.215)	(0.293)
Mean of outcome	0.43	0.45	0.89	0.89
SD of outcome	0.68	0.69	1.04	1.05
N	2,995	1,409	1,375	176

This table reports coefficient estimates from the estimation of the main specification where the primary regressor has been transformed into a vector of indicators for differing levels of cumulative exposure intensity. Coefficients are from the estimation of this modified form of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Standard errors are two-way clustered by constituency and district. Significant at * <.1, ** <.05, and *** <.01.

years, three to five years, six to seven years, and eight years or more. In both state legislative assemblies and parliamentary elections, we see that the majority of the effect comes from areas with greater than five years (i.e., more than one election cycle) of reservation. In appendix table A8, I estimate a district fixed-effects specification, the results of which correspond in direction to the main effects above, although smaller and statistically insignificant.

VII. Representation Effects

A. *Winning, Election Finishes, and Effects on Electoral Competition*

Table 10 estimates changes in the share of votes garnered by female candidates and in the (unconditional) probability of a woman winning the election or finishing in either the top five finishers or the top 30% of finishers. Column 1 includes the change in the share of candidates who are female for a comparable reference. Column 2 shows that female candidates capture at least a proportionate share of votes (although imprecisely estimated). This vote share effect is likely to be quite small if the marginal candidate garners a proportional (average) share of votes. Using the parliamentary elections as an example, for an additional female candidate in a pool of thirteen candidates, the fraction of votes going to female candidates would increase by at most 1/14, or 0.07. However, since the average effect on candidates is 0.041 individuals, the expected change in the vote share then becomes quite small: $0.041 \times (1/14) = 0.00292$. Despite the assumption regarding proportionality (which may be strong for the marginal female candidate), this is not far from the magnitude estimated in

TABLE 10.—EFFECTS OF POLICY EXPOSURE ON FEMALE CANDIDATES' ELECTION FINISHES

Outcome:	Female Share of Candidates (1)	Female Vote Share (2)	Female Winner (3)	Female Finish in: Top Five (4)	Female Finish in: Top 30% (5)
A. State assemblies, full sample					
Years reserved	0.00307**	0.00200	0.00110	0.01096	0.00239
	(0.00154)	(0.00218)	(0.00268)	(0.00793)	(0.00577)
B. State assemblies, overlapping sample					
Years reserved	0.00339**	0.00245	0.00106	0.00432	0.00142
	(0.00168)	(0.00237)	(0.00287)	(0.00635)	(0.00528)
C. Parliament, full sample					
Years reserved	0.00274	0.00546	0.01147	0.01541	0.01705
	(0.00204)	(0.00509)	(0.01010)	(0.01221)	(0.01161)
D. Parliament, overlapping sample					
Years reserved	0.00480*	0.00993	0.01443	0.02854*	0.03045*
	(0.00256)	(0.00724)	(0.01503)	(0.01591)	(0.01624)

This table reports coefficient estimates of the effect of an additional year of exposure to female leaders on segments of elections' outcome finishes. Coefficients are from the estimation of equation (1), estimated via OLS. Sample sizes across panels A–D are 2,995, 1,409, 1,375, and 176, respectively. All specifications in all panels include a vector of state fixed effects. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

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TABLE 11.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON PARLIAMENTARY PARTY COMPETITION ($N = 1{,}375$)

	Female Candidates, Major Party (1)	Female Candidates, Minor Party (2)	Female Candidates, Independent (3)	Vote Share, Major Party Candidate (4)
Years reserved	0.014	0.005	0.022**	−0.024
	(0.017)	(0.012)	(0.011)	(0.016)
Mean of outcome	0.30	0.24	0.31	0.47
SD of outcome	0.53	0.50	0.62	0.60

This table reports coefficient estimates of the effect of an additional year of exposure to female leaders on constituency-level metrics of female candidates by party type. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

table 10. Using the figures for state elections, the expected vote share increase ($0.015 \times (1/9) = 0.00166$) is quite similar to the estimated value (0.00200). From this, we can conclude that the effects estimated are in fact consistent with marginal female candidates' garnering an approximately proportional vote share.

Column 3 shows that there is no distinguishable increase in the probability of electing a female candidate in either state or national legislatures. This is important to note in the context of the findings of Bhalotra et al. (2018), who show that political experience causes an increase in female candidacies through repeat candidates. In the present situation, the effect of the quota policy could be compounded with positive incumbency effects if the female candidates who contested as a result of quotas also are likely to win elections. This is not the case, however, and provides one reason for the impermanence of cross-level effects of local quotas.

Focusing only on whether women win elections may understate changes in the viability of female candidates. To address this, I construct alternative measures of whether a female candidate finished in either the top five candidates or in the top 30% of candidates (to adjust for the size of the candidate pool). Columns 4 and 5 contain the results of these estimations and indicate there are no substantial effects on female candidates finishing in the top of the distribution of finishers in the elections in which they compete, suggesting the additional female candidates induced by the policy receive relatively low vote shares in the elections they contest, finishing no better or worse than the average candidate. However, there is some indication of a higher probability of a female candidate having a competitive finish (columns 4 and 5) in the parliament in highly overlapping areas (panel D). Appendix figure A3 shows the progression of these estimates when restricting the sample to more similarly overlapping areas.

The investigation into individual finishes does not allow for indirect effects of additional candidates on the election contest overall, given that marginal candidates may still cause changes in voting behavior of particular blocs to which they are particularly connected. In table 11, I establish several facts about the characteristics of candidates who respond to policy exposure via parliamentary candidacy. To do this, I segment candidates into whether they ran for office as independents or as part of a major or minor party, and then look at the effect of policy exposure on the share of votes won by major party candidates. This establishes important points about the candidates who run for parliament in constituencies longer exposed to quotas: they largely run as independents rather than as party candidates (columns 1–3), and policy exposure may reduce the vote share going to major party candidates (column 4).[9] These marginal candidates may thus indirectly affect electoral outcomes through independent candidacies that disproportionately reduce the vote share going to major party candidates, although this effect is not statistically significant.

TABLE 12.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON THE ELECTION OF FEMALE CANDIDATES CONDITIONAL ON PARTY TYPE

Outcome:	Female Winner (1)	Female Finish In: Top Five (2)	Female Finish In: Top 30% (3)
A. Female candidates > 0			
Years reserved	0.016	0.011	0.013
	(0.015)	(0.012)	(0.013)
B. Major party female candidates > 0			
Years reserved	0.034*	0.023*	0.023
	(0.021)	(0.012)	(0.017)
C. Minor party female candidates > 0			
Years reserved	0.022	0.021	0.046**
	(0.022)	(0.023)	(0.023)
D. Independent female candidates > 0			
Years reserved	−0.032*	−0.012	−0.032*
	(0.017)	(0.016)	(0.019)

This table reports coefficient estimates of the effect of an additional year of reservation on an indicator for a female candidate winning the election. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

I next estimate the effect of quota exposure on the probability of a female candidate winning conditional on the presence of different types of female candidates. That is, does quota exposure have a stronger effect on the probability of a female candidate winning when the candidate has been incorporated into a major party? Table 12 presents the estimation of the specifications in columns 3, 4, and 5 of table 8 conditional on the presence of a female candidate (panel A), the presence of a major-party female candidate (panel B), the presence of a minor-party female candidate (panel C), and the presence of

[9]There were four major party candidates in a constituency, on average, suggesting the expected vote share loss for major party candidates would be approximately 10% of the magnitude in column 4.

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TABLE 13.—EFFECT OF EXPOSURE TO FEMALE LEADERS ON VOTER MOTIVATION AND TURNOUT

Level:	Parliament			Assemblies		
	Voter Turnout (1)	Female Share Voters (2)	Female Voter Turnout (3)	Voter Turnout (4)	Female Share Voters (5)	Female Voter Turnout (6)
Years reserved	0.00073	0.00037	0.00082	0.00302*	−0.00004	0.00296
	(0.00158)	(0.00052)	(0.00183)	(0.00170)	(0.00031)	(0.00190)
N	1,375	1,375	1,375	2,995	2,993	2,993
Mean of outcome	0.61	0.46	0.59	0.67	0.49	0.65
SD of outcome	0.15	0.04	0.16	0.13	0.02	0.13

This table reports coefficient estimates of the effect of an additional year of reservation on constituency-level metrics of voter turnout outcomes. Coefficients are from the estimation of equation (1), estimated with OLS. All specifications contain a vector of state fixed effects and an unreported constant term. Heteroskedasticity-consistent standard errors clustered by district and constituency in parentheses. Significant at * <.1, ** <.05, and *** <.01.

an independent female candidate (panel D). When abstracting from selection concerns, a clear pattern emerges across panels: conditional on candidacy, women who are incorporated into major parties increase their chances of election when their constituency has previously had greater exposure to quotas, whereas this is the opposite for independent candidates. These results parallel the findings among municipal elections of Bhavnani (2009) by showing that the most likely conduit to achieve follow-on representation effects for women after a quota regime is through the concurrent incorporation of female candidates into major parties. Conditioning the sample on the presence of a female candidate running on a major party ticket does introduce concerns about results being driven by an endogenously determined sample arising from parties' candidate selection processes, however. Absent selection, this would suggest that representation effects of quota exposure can only be realized in combination with major party support. If these results arise due to selection, this could instead be taken as evidence of a change in major parties' assessment of female candidate viability in longer exposed areas, suggesting a different form of institutional change.

B. *Effects on Voter Turnout*

I use measures of voter turnout in parliamentary and assembly elections to investigate whether there is evidence that female candidacy is increased indirectly by expectations of enhanced voter turnout by groups potentially more likely to vote for female candidates. For both parliamentary and state assembly constituencies, voter turnout measures are available separately by sex. Table 13 reports point estimates from the same empirical framework as above. While there is a precisely estimated zero effect of female leaders on measures of voter turnout in parliamentary elections, there is some increase in turnout in state assembly elections (column 4), which appears to be due largely to an increase in female voter turnout (column 6). Effect magnitudes imply that an additional two terms of exposure to the quota reservation increases female voter turnout by a fraction of .03, an increase of 5% over the mean, or approximately .23 standard deviation, suggesting some potential for quota-based leadership to encourage the political participation of female voters.

VIII. Conclusion

This paper provides evidence that a quota policy for women in local government increased the candidacy, but not representation, of women in higher offices. Using a natural experiment in which one-third of leadership seats in local government were reserved for women across election cycles, I identify a causal response to these seat quotas among women contesting seats in later elections for state legislative bodies and the national parliament. This suggests additional, longer-term effects of quotas on political dynamics and effects outside the particular bodies to which the quotas were applied. Estimate magnitudes imply these quotas were responsible for a majority of the increase in female candidates in state legislature and parliamentary elections since the policy went into effect.

The cross-level effects of quotas in politics were linked to two distinct groups of individuals. These were candidates who had either gained previous experience in local government due to the quotas or individuals who were not direct beneficiaries of the policy but were previous candidates who continued contesting for office in areas that had greater exposure to local female leaders. There was no overall increase in the probability of electing a woman despite the increase in female candidates, although there is some indication of a higher likelihood of a competitive finish by female candidates in longer-exposed areas. There is also suggestive evidence that female representation increases when women are able to run on major party tickets. Effects on the extensive margin of voting are limited, although the additional female candidates disproportionately reduce the vote share going to major party candidates. If a policy goal is to increase substantive representation throughout politics, quotas in local government have distinct, but limited, effects on representation in higher levels of the political structure.

The findings have implications for understanding how quotas can affect later candidacy for higher-level positions in the occupation in which they operate. India's one-third quota policy for women in local politics has increased the substantive representation of women in local government over the previous two decades. The relative success of these quotas in local government, however, has not yet translated into an increase in representation by women in higher levels of government. Follow-on representation effects are not automatic;

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they may be accessible only in environments where candidates can avail sufficient resources needed to compete for higher-level positions.

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